

Matthew Yu

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SKILLS

Languages Python, Java, C, C++, C#, TypeScript/JavaScript, SQL, T-SQL, Lua, Racket, Ruby (Ruby on Rails), OpensCAD
Web React, React Native Next, Express, ASP.NET, Flask, Spring Boot
Data/ML , Relational Databases (Mysql, Ms Server), Web Scraping, Tensorflow (Keras), Pytorch, OpenCV, Parser
Tools Git, Bash, Postman, Vim, Docker, Kubernetes

EXPERIENCE

The EPCM Group Global – Software Developer Intern

June 2026 – Present

Technologies: React, C#, ASP.NET, Docker, Minimal Apis, IIS, MS Server

Oakville, ON

- Created and deployed a full internal tool to speed up the proposal and quoting process
- Created and deployed Automated Console apps that fetches commodity process running on SQL agent jobs
- Regular communication to important users to ensure full satisfaction.

Wedding Strings International – Mobile Software Developer

June 2026 – September

Technologies: Javascript, Supabase, React Native, Expo

Toronto, ON

- Creating a personalized internal app to streamline the process of booking, scheduling, and keeping track of musicians playing at various weddings across the world
- Used by hundreds of musicians

Reality Labs Design Team – Software Engineer (Humanoid Team)

Sept 2025 – April 2026

Technologies: JavaScript, C#, Express, WebSockets, Unity

Waterloo, ON

- Developed backend services for a **Unity VR** application using **Express** and **WebSockets**, achieving **sub-50 ms** end-to-end interaction latency from VR Input to App

Project: Empower – Software Engineer (Backend)

Aug 2024 – Aug 2025

Technologies: Python, Flask, Beautiful Soup

Remote

- Built reliable **Flask** backend for app that collects student opportunities from external web sources using **Beautiful Soup**.

Mu Systems Corporation – Software Engineer (Backend)

Jul 2024 – Aug 2024

Technologies: Java, Spring Boot, Python, Beautiful Soup, Zentao

Remote

- Developed backend features for internal tools at an **e-commerce startup**, including **route optimization** for delivery drivers, item search using **Elasticsearch**, and automated competitor price collection via **web scraping**.

AWARDS & QUALIFICATIONS

Competitive Programming — Codeforces (Top 10%), DMOJ (Top 700), USACO Silver

2021 – Present

11x Hackathon Participant (2x Winner)

2024 - Present

3rd Place — Forester Pitch Challenge at UofT Hacks

2026

1st Place — PTC Hacks

2024

SHAD Program — Best Design Award for Website Creation

2024

PROJECTS

Bricked — Constraint-Aware LEGO Design Platform



Next.js, React, Three.js, OpenCV, OpenSCAD, FastAPI, LangGraph, WebAssembly, Webworker, Manifold

- Digitized real LEGO inventory via **computer vision** for **accurate** 3D reconstruction
- Generated LEGO builds using a **multi-agent AI pipeline (LangGraph)** for **quality** and based on scanned bricks
- Built & Optimized an OpenSCAD compiler (WASM + Manifold), reducing render times
- Delivered interactive, step-by-step 3D build instructions with per-brick customization in DIY **Solidworks-like** platform

Focus Friend — EEG-Based Focus Detection System



React Native, TypeScript, C++, Hardware (Arduino / Operational Amplifiers)

- Constructed a mobile app that streams **EEG-derived signals** and **visualizes** focus vs. fatigue in real time
- Implemented a data pipeline for **preprocessing** noisy sensor input and running on-device/in-app inference
- Integrated hardware acquisition with a **React Native UI** and **Bluetooth**, to deliver feedback loops for users

EDUCATION

University of Waterloo

Present

Bachelor of Computer Science (Honours)

Waterloo, ON

GPA: 4.0 • **Clubs:** CS, DS • **Advanced Math** 145 & CS 146 • VP of Operations, Residence Council

More projects and details: matthewyu.tech